

Hongpei Li

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EDUCATION

Northwestern University Ph.D. in Industrial Engineering and Management Sciences (IEMS)	2026 - 2032 (Expected) Evanston, IL
Shanghai University of Finance and Economics Bachelor of Engineering, Pilot Class of Interdisciplinary Sciences GPA 3.77/4.0 Average Score 89.55/100 Major GPA 3.87 Major Average Score 91.48/100 Selected Coursework: Advanced Operation Research, Linear and Nonlinear Programming, Probability, Numerical Optimization, Mathematical Analysis, Linear Algebra, Data Structures & Algorithms, Machine Learning & Deep Learning. Honors & Awards: - Honor Bachelor Degree, 2025. (Top 5% in Major) - Honor of Excellence Graduation Thesis, 2025. (Top 3% in Major) - Shanghai Municipal Bronze Award in the Internet+ University Student Innovation and Entrepreneurship Competition. (0.13%)	2021 - 2025 Shanghai, China

WORK EXPERIENCE

LLM & OPT Researcher QingTian Lab, Cardinal Operations	Feb.2025 - Present Shanghai, China
- LLM Systems: Research on infrastructure optimization for large-scale LLM training. - Optimization: Research on GPU-accelerated parallel optimization algorithms.	

RESEARCH INTERESTS

Mathematical Optimization, Large-Scale Optimization, AI for Optimization, Language Model Architectures, ML System

PUBLICATIONS

LARGE-SCALE OPTIMIZATION

- [P1] A Restarted Primal-Dual Hybrid Conjugate Gradient Method for Large-Scale Quadratic Programming (**Accepted by INFORMS Journal on Computing**)
Y. Huang, W. Zhang, **H. Li**, D. Ge, H. Liu, and Y. Ye. (2024). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))
- [P2] PDHCG-II: An Enhanced Version of PDHCG for Large-Scale Convex QP
H. Li, Y. Huang, H. Liu, D. Ge and Y. Ye (2026). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))
- [P3] A Scalable First-Order Method for Large-Scale Competitive Market Equilibrium Computation (**Submitted to INFORMS Journal on Computing**)
H. Liu, Y. Huang, **H. Li**, D. Ge and Y. Ye (2025). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))
- [P4] D-PDLP: Scaling PDLP to Distributed Multi-GPU Systems
H. Li, Y. Huang, H. Liu, D. Ge and Y. Ye (2026). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))

DEEP LEARNING APPLICATION

- [P5] A Learning and Searching Method for Integrated Processing Planning and Scheduling (**Submitted to Production and Operations Management**)
H. Li, H. Zhang, Z. He, Y. Jia, B. Jiang, X. Huang, and D. Ge. (2024). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))
- [P6] BenLOC: A Benchmark for Learning to Configure MIP Optimizers
H. Li, Z. He, Y. Wang, S. Pu, Q. Deng, W. Tu, and D. Ge. (2025). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))
- [P7] FMIP: Joint Continuous-Integer Flow for Mixed-Integer Linear Programming (**Accepted by ICLR 2026**)
H. Li, H. Yuan, H. Zhang, J. Lin, D. Ge, M. Wang and Y. Ye (2025). *arXiv preprint* ([\[Paper\]](#))([\[Code\]](#))

ML SYSTEM

- [P8] OptPipe: Memory- and Scheduling-Optimized Pipeline Parallelism for LLM Training
H. Li, H. Zhang, H. Liu, D. Ge and Y. Ye (2025). *arXiv preprint* ([\[Paper\]](#))
- [P9] OSDN: Improving Delta Rule with Provable Online Preconditioning in Linear Attention
C. Zhou, **H. Li**, Y. Liu, J. Lin, D. Ge and Y. Ye (2026). *arXiv preprint* ([\[Paper\]](#))

SELECTED RESEARCH EXPERIENCE

LARGE-SCALE OPTIMIZATION

Improvement of Primal-Dual Hybrid Gradient Method For Large-Scale Linear Programming Oct.2025 - Present

Adviser: Prof. Haihao Lu Massachusetts Institute of Technology

Further improvement the efficiency and design post-processing phase of cuPDL Px, a GPU-accelerated primal-dual hybrid gradient method for large-scale linear programming problems.

- Develop fused GPU kernels, specified for the features of problem instances, to further improve the efficiency of cuPDL Px.
- Design a post-processing phase to further improve the solution quality of cuPDL Px after the major PDHG iterations.

GPU-Accelerated First-Order Method for the Robotic Control via SOS Relaxation Jun.2025 - Present

Adviser: Prof. Yinyu Ye, Prof. Huikang Liu Stanford University

Developing a high-performance first-order solver for large-scale Semidefinite Programming (SDP) arising from Sum-of-Squares (SOS) relaxations in robotic control and trajectory planning.

- Implement whole algorithm framework and solver components.
- Design efficient multi-GPU PSD projection algorithm and other fused GPU kernels.

GPU-Accelerated First-Order Method for Large-Scale Fisher Equilibrium Problems [P3] Nov.2024 - Apr.2025

Adviser: Prof. Yinyu Ye, Prof. Huikang Liu Stanford University

An efficient and GPU-accelerated first order method for large-scale Fisher equilibrium problems.

- Implement efficient GPU kernels for efficiency improvement.
- Develop a CUDA-C based solver for large-scale Fisher equilibrium problems.

Primal-Dual Hybrid Conjugate Gradient Method (PDHCG) [P1] Apr.2024 - Oct.2024

Adviser: Prof. Yinyu Ye, Prof. Dongdong Ge, Prof. Huikang Liu Stanford University, Shanghai Jiao Tong University

An efficient and GPU-accelerated first order method for large-scale quadratic programming problems.

- Implement GPU version and low-rank acceleration of PDHCG solver.
- Implement computational techniques on GPU, including asynchronous computation, fused kernels, memory access optimization.

DEEP LEARNING APPLICATION

Generative Models for Linear Programming&Mixed Integer Programming [P7] Sep.2024 - May.2025

Adviser: Prof. Mengdi Wang, Prof. Yinyu Ye Princeton University, Stanford University

Use generative models to accelerate the solving of linear programming (LP) and mixed-integer programming (MIP) problems.

- Build a framework for generating high qualified solution of LPs and MILPs using flow matching.
- Design a joint continuous-integer flow for MILP problems and a optimality-aware training-free guidance strategy.

Deep Reinforcement Learning (DRL) for Scheduling Problems [P5] Mar.2024 - Aug.2024

Adviser: Prof. Dongdong Ge, Prof. Bo Jiang Shanghai Jiao Tong University, SUFE, Cardinal Optimizer

Use DRL to solve the integrated process planning and scheduling problem, a kind of realistic and difficult scheduling problem.

- Implement a GPU-based simulation environment for training and Design a novel graph representation of the problem, a dense reward function and a real-time action space reduction method.
- Design a pruning strategy to improve learning-guided searching method.

Machine Learning for MIP Optimizer Configuration [P6] Dec.2023 - Oct.2024

Adviser: Prof. Dongdong Ge Shanghai Jiao Tong University

Use machine learning to help configure MIP optimizers for better performance.

- Build a whole framework from dataset generation, feature engineering, model training to deployment.
- Design GNNs to predict the best configuration of MIP optimizers.

ML SYSTEM

Improve Pipeline Parallelsim using Mathematic Programming [P8] Mar.2025 - Present

Adviser: Prof. Yinyu Ye Stanford University

Use mathematic programming to optimize the memory and scheduling of pipeline parallelism in LLM training.

- Implement a framework based on Megatron-LM, which including building and solving a mixed-integer programming model.
- Design practical techniques to reduce required solving time and improve the solution quality.

VOLUNTEER EXPERIENCE

Shanghai University of Finance and Economics Shanghai, China

- Teaching Assistant: Advanced Operations Research (Spring '24), Object-Oriented Analysis & Design (Fall '23), Programming Peer Tutor (Fall '22).
- Service & Volunteering: Member of Students' Union (2021-2022); Campus Return Recruitment Volunteer (Winter 2021).